RAHAT AHMED JOBU

PhD Applicant: Software Engineering, Machine Learning & NLP

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RESEARCH PROFILE

CSE student at BRAC University working toward a PhD in software engineering, machine learning and NLP. My research centers on detecting systems designed to look trustworthy but aren't: fraudulent text and job postings, malicious software packages, silent API breaking changes, and LLM agents that act on scams they can recognize. A defining practice across this work is adversarial self-evaluation: multiple rounds of self-directed critique targeting train/test leakage, confounded baselines and unverified claims, applied to my own results before they reach peer review.

EDUCATION

BRAC University, Dhaka Jan 2022 – Expected Jan 2027
B.Sc. in Computer Science and Engineering (transferring to B.Sc. in Computer Science, Fall 2026)

Higher Secondary Certificate (HSC), Science, Shaheed Police Smrity College GPA 5.00/5.00
Secondary School Certificate (SSC), Science, Green Heart School GPA 5.00/5.00

PUBLICATIONS

Published (Peer-Reviewed Journal Articles)

1. **Jobu, R. A.**, Baral, A., Rahman, S. M. S., & Probor, M. N. (2026). An explainable, cost-aware hybrid framework for fraudulent online job posting detection. *Array* (Elsevier, Q1), 31, 101191. doi:10.1016/j.array.2026.101191. First and corresponding author.
2. **Jobu, R. A.** (2026). KnowGuard: An Explainable, Outcome-Validated Decision-Support System for Knowledge-Concentration Risk Mitigation in Software Projects. *Results in Engineering* (Elsevier, Q1). doi:10.1016/j.rineng.2026.113833. Single author.

Under Review and in Revision

3. **Jobu, R. A.** How Much Does Data Leakage Inflate ML4SE Benchmarks? A Cross-Task Empirical Audit. *Under review, ACM International Conference on the Foundations of Software Engineering (FSE 2027), Research Track.*
4. **Jobu, R. A.** Engineering a Lightweight, Cross-Generator-Robust Pipeline for Bengali AI-Paraphrase Detection: A Leakage- and Artifact-Aware Evaluation Process. *Results in Engineering (Elsevier, Q1); major revision.*
5. **Jobu, R. A.** PyPIGuard: A Lightweight, Explainable Machine Learning Tool for Pre-Install Detection of Malicious PyPI Packages. *Under review, SoftwareX (Elsevier).*
6. **Jobu, R. A.** Does Naive Cross-Validation Inflate Software Energy-Regression Prediction? A Leakage-Aware, Statistically-Gated, Cross-Project Empirical Study. *Under review, Information and Software Technology (Elsevier, Q1).*
7. **Jobu, R. A.** Knowing Is Not Doing: Open-Weight LLM Job-Search Agents Act on Employment Scams Their Own Models Flag. *Submitted to The Web Conference (WWW 2027), Security and Privacy Track.*

Preprints of items 2 and 4–6 are available on SSRN.

EXPERIENCE

Software Engineer, ASG SHOP (AI Solutions Garage) Jul 2026 – Present

- Design, develop and maintain features across the company's software products; work with cross-functional teams to turn business requirements into technical solutions.
- Part of the engineering team maintaining the infrastructure behind Apar's Classroom, an EdTech platform serving 1M+ students in Bangladesh, ranked #2 globally on TIME's World's EdTech Rising Stars 2026 list.

Software Engineering Intern, ASG SHOP (AI Solutions Garage) Jan 2026 – Jun 2026

- Designed, developed and tested web application modules under senior engineers; took part in code reviews, stand-ups and sprint planning.

Software Engineer, IOOTE Dec 2024 – Sep 2025

- Developed full-stack web features using React, Node.js and MongoDB for tech-gadget products.
- Built REST APIs and integrated front-end and back-end services with designers and product stakeholders.

SELECTED PROJECTS

- **Job Fraud Detector:** Hybrid SVM + DistilBERT pipeline for fraudulent job posting detection on EMSCAD; implementation behind the *Array* paper. github.com/rahat239/job-fraud
- **PyPIGuard:** ML tool for pre-install detection of malicious PyPI packages, using cluster-disjoint evaluation to remove near-duplicate leakage. github.com/rahat239/PyPIguard-Research
- **Techra:** PC builder and e-commerce platform (MERN stack). github.com/rahat239/TechRA
- **J.A.R.V.I.S.:** Personal macOS AI assistant (~5,500 lines of Python) with voice, gesture and face recognition.

ENTREPRENEURSHIP

Founder, MeowMatchBD meowmatchbd.com

Full-stack platform for cat adoption, live rescue and nearby veterinarian discovery in Bangladesh; revenue from paid sell postings. 1,200+ registered users and 218 cats adopted to date. Responsible for end-to-end product development and operations.

HONORS AND ACHIEVEMENTS

- Selected among the top 8 projects for the industry-expert showcase, CSE471 (System Analysis and Design), BRAC University, Summer 2025; evaluated by experts from Brain Station 23, Optimizely and Cefalo Bangladesh; certificate of recognition.
- District Champion, Painting Competition; 2nd Place, Regional Painting Competition.

OTHER

- **Book:** *Predicting Silent Behavioral Drift Severity: From Default-Argument Breaking Changes in Python Machine Learning Libraries.* Eliva Press.
- Former Executive, BRAC University Computer Club.

TECHNICAL SKILLS

Machine Learning / NLP: Python, PyTorch, scikit-learn, Transformers; statistical evaluation (cluster-disjoint validation, bootstrap resampling)
Software: JavaScript, React, Node.js, Express.js, MongoDB, C++, Arduino, Git